

Cocoro Wachi

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Summary

Computer Science major at Milwaukee School of Engineering with minors in Mathematics and UX Design, seeking an internship in applied machine learning research and model development. Experience includes ML research, time series analysis and visualization, cloud computing & architecture with AWS. Conducted applied ML research in urban hydrology using 12+ years of sensor data and presented at an international conference. Skilled in translating complex models into user-focused analytical tools. Fluent in Japanese and English.

Education

Bachelor of Science, Computer Science. Mathematics Minor, User Experience Design Minor.

Milwaukee School of Engineering | GPA 3.4 | Expected Graduation May 2027

Technical Skills

Programming & Dev: Python, SQL, C/C++, Java, JavaScript, AWS S3, Git, GitLab, Bash/Shell Scripting, CI/CD

Data Science: NumPy, Pandas, SciPy, TensorFlow, Scikit-learn, Hyperparameter Optimization, Model Validation

Mathematics: Abstract Algebra, Finite Group Theory, Linear Algebra, Probability & Statistics, Complex Analysis

Internship Experience

RJN Group Inc., Downers Grove, IL

Summer 2025

Full-Stack Software Engineering and Hydraulic Modeling Intern

- Developed a full-stack time series modeling dashboard with real-time, interactive data editing & visualization features using Python, Git, Streamlit, and AWS S3, improving engineers' modeling productivity by ~20%.
- Researched and built an optimization tool with cloud-based data versioning for collaborative workflows.
- Cross-functional feature demos for engineers, refining backlog priorities for short and long-term user needs.

Research Experience

ML Model Development and Analysis for Urban Hydrology

Jan. 2025 - Present

Undergraduate Research, Milwaukee School of Engineering, Dr. William Gonwa P.E., Advisor

- Led exploratory and statistical analysis on 12+ years of multi-site sensor data (50+ gauges), identifying structural model biases and regime-dependent performance failures.
- Developed a ML parameter optimization tool that addressed model complexity and interpretability.
- Presented at ICWMM 2026 and a planned research publication from JWMM as primary author.

NLP-Based Unstructured PDF Ingestion and Text Analysis Pipeline | EPlanExam

Jan. 2026 - Present

- Collaborated with QA engineers to define PBIs for an NLP-driven data analysis system and software.
- Applied NLP methods (tokenization, vectorization, document embeddings) across 17,000+ regulatory files.
- Designed a scalable preprocessing pipeline to support downstream analysis and visualization of insights.

Employment History

Math & Physics Tutor | MSOE Student Employment | 7 hours per week

Sept. 2024 - Present

Pep Band Performer | MSOE Pep Band | 7 - 10 hours per week

Sept. 2023 - Present

Server, Host | Shabu-Yo Skylark | 10 - 40 hours per week

Oct. 2022 - Jan. 2025

Custodial, Groundskeeping | MSOE Student Employment | 7 hours per week

Sept. 2023 - Dec. 2023

Leadership & Awards

President, Co-Captain | AISC Student Steel Bridge Competition

Sept. 2024 - Present

MSOE Dean's List (5x) | MSOE Dean's Office

Fall 2023 - Present

Principal Trumpet | MSOE Concert Band & Jazz Band

Sept. 2023 - Present

Food Preparation, Distribution | MLK Day Open Door Cafe Volunteer

Jan. 2025

Performer | Wisconsin Concert Band Association Intercollegiate Honor Band

Jan. 2025

Performer | Madison Scouts Drum & Bugle Corps

Summer 2024